Run report: H2-glue-run-20261018T2240Z

Descriptive record of one recorded run of examples/n2_routes.rs, modes gluecheck and glue (Addendum H2/H3), on a RunPod CPU pod under `research/weil_descent_angles_20261018/PROTOCOL.md` (Addendum A and Amendments B–G and Addendum H, each committed before the run it governs). It lists what ran, on what host, and what the tool recorded, file by file. Interpretation is in the study's `REPORT.md`. Costs are counted operations (F_p multiplications and inversions, group operations, native operations), never wall-clock; the only times are the tool's own seconds. Isolation: none earned (a shared cloud container, or a RunPod CPU pod whose lscpu is in host.txt); this affects only the seconds, since every recorded value is an exact count or a counted operation total.

env.txt

```
commit 0bd2a0f5c0b0f0005d57850c12466fa82b414c59 (fetched by the pod from raw.githubusercontent.com, sha256-checked)
cmd: rustc --edition 2021 -O examples/n2_routes.rs -o n2 && ./n2 "$OUT" gluecheck && N2_THREADS=16 ./n2 "$OUT" glue
host: RunPod CPU pod p22sy8golm3xb0 (h2-glue), 16 vCPU, flavors cpu5c,cpu3c, image rust:1-bookworm, launched 2026-10-18T22:24:37Z
cpu: AMD EPYC 4564P 16-Core Processor
rustc 1.99.1 (58cf8f9c3 2026-10-13)
runner: research/weil_descent_angles_20261018/runpod/podrun.py
post-run (ffabb7cd8): glue_summary.txt re-derived from glue.csv by `n2_routes <dir> gluesummarize` (adds the pricing band with the Legendre split test at one I, Addendum H3); glue.csv unchanged.
```

glue_checks.txt

```
G p=131 N=17209 h=1 ell=17209 t=-47: t2=sigma(s2) true lambda^2 in F_p true lambda in F_p true | twist true | G1 #C(F_p)=132 (p+1=132) #C(F_p2)=17256 (p^2+1-2t=17256) | G2 assoc true inverse
true N*A=0 true | G3 hom true nonzero true ell*tau=0 true
G p=137 N=18911 h=1 ell=18911 t=-141: t2=sigma(s2) true lambda^2 in F_p true lambda in F_p true | twist true | G1 #C(F_p)=138 (p+1=138) #C(F_p2)=19052 (p^2+1-2t=19052) | G2 assoc true inverse
true N*A=0 true | G3 hom true nonzero true ell*tau=0 true
G p=131 N=17132 h=4 ell=4283 t=30: t2=sigma(s2) true lambda^2 in F_p true lambda in F_p true | twist false | G1 #C(F_p)=132 (p+1=132) #C(F_p2)=17102 (p^2+1-2t=17102) | G2 assoc true inverse true
N*A=0 true | G3 hom true nonzero true ell*tau=0 true
G p=139 N=19204 h=4 ell=4801 t=118: t2=sigma(s2) true lambda^2 in F_p true lambda in F_p true | twist false | G1 #C(F_p)=140 (p+1=140) #C(F_p2)=19086 (p^2+1-2t=19086) | G2 assoc true inverse
true N*A=0 degenerate | G3 hom true nonzero true ell*tau=0 true
G p=277 N=76603 h=1 ell=76603 t=127: t2=sigma(s2) true lambda^2 in F_p true lambda in F_p true | twist false | G1 #C(F_p)=278 (p+1=278) #C(F_p2)=76476 (p^2+1-2t=76476) | G2 assoc true inverse
true N*A=0 degenerate | G3 hom true nonzero true ell*tau=0 true
G p=257 N=66301 h=1 ell=66301 t=-251: t2=sigma(s2) true lambda^2 in F_p true lambda in F_p true | twist false | G1 #C(F_p)=258 (p+1=258) #C(F_p2)=66552 (p^2+1-2t=66552) | G2 assoc true inverse
true N*A=0 true | G3 hom true nonzero true ell*tau=0 degenerate
G p=271 N=73252 h=4 ell=18313 t=190: t2=sigma(s2) true lambda^2 in F_p true lambda in F_p true | twist false | G1 #C(F_p)=272 (p+1=272) #C(F_p2)=73062 (p^2+1-2t=73062) | G2 assoc true inverse
true N*A=0 true | G3 hom true nonzero true ell*tau=0 true
G p=257 N=66212 h=4 ell=16553 t=-162: t2=sigma(s2) true lambda^2 in F_p true lambda in F_p true | twist false | G1 #C(F_p)=258 (p+1=258) #C(F_p2)=66374 (p^2+1-2t=66374) | G2 assoc true inverse
true N*A=0 true | G3 hom true nonzero true ell*tau=0 true
G p=1033 N=1065949 h=1 ell=1065949 t=1141: t2=sigma(s2) true lambda^2 in F_p true lambda in F_p true | twist false | G1 #C(F_p)=1034 (p+1=1034) #C(F_p2)=- (p^2+1-2t=1064808) | G2 assoc true
inverse true N*A=0 true | G3 hom true nonzero true ell*tau=0 true
G p=1109 N=1231631 h=1 ell=1231631 t=-1749: t2=sigma(s2) true lambda^2 in F_p true lambda in F_p true | twist false | G1 #C(F_p)=1110 (p+1=1110) #C(F_p2)=- (p^2+1-2t=1233380) | G2 assoc true
inverse true N*A=0 true | G3 hom true nonzero true ell*tau=0 true
G p=1097 N=1201292 h=4 ell=300323 t=2118: t2=sigma(s2) true lambda^2 in F_p true lambda in F_p true | twist false | G1 #C(F_p)=1098 (p+1=1098) #C(F_p2)=- (p^2+1-2t=1199174) | G2 assoc true
inverse true N*A=0 true | G3 hom true nonzero true ell*tau=0 true
G p=1039 N=1079564 h=4 ell=269891 t=-42: t2=sigma(s2) true lambda^2 in F_p true lambda in F_p true | twist false | G1 #C(F_p)=1040 (p+1=1040) #C(F_p2)=- (p^2+1-2t=1079606) | G2 assoc true
inverse true N*A=0 true | G3 hom true nonzero true ell*tau=0 true
G p=4507 N=20306837 h=1 ell=20306837 t=6213: t2=sigma(s2) true lambda^2 in F_p true lambda in F_p true | twist false | G1 #C(F_p)=4508 (p+1=4508) #C(F_p2)=- (p^2+1-2t=20300624) | G2 assoc true
inverse true N*A=0 true | G3 hom true nonzero true ell*tau=0 true
G p=4457 N=19863127 h=1 ell=19863127 t=1723: t2=sigma(s2) true lambda^2 in F_p true lambda in F_p true | twist false | G1 #C(F_p)=4458 (p+1=4458) #C(F_p2)=- (p^2+1-2t=19861404) | G2 assoc true
inverse true N*A=0 true | G3 hom true nonzero true ell*tau=0 true
G p=4549 N=20695252 h=4 ell=5173813 t=-1850: t2=sigma(s2) true lambda^2 in F_p true lambda in F_p true | twist false | G1 #C(F_p)=4550 (p+1=4550) #C(F_p2)=- (p^2+1-2t=20697102) | G2 assoc true
inverse true N*A=0 true | G3 hom true nonzero true ell*tau=0 true
G p=4133 N=17083268 h=4 ell=4270817 t=-1578: t2=sigma(s2) true lambda^2 in F_p true lambda in F_p true | twist false | G1 #C(F_p)=4134 (p+1=4134) #C(F_p2)=- (p^2+1-2t=17084846) | G2 assoc true
inverse true N*A=0 true | G3 hom true nonzero true ell*tau=0 true
G5 p=1063: fast = generic on 800/800 pairs (0 degenerate) | add fast 41.0M+1.00I generic 112.8M+4.00I | double fast 51.0M+1.00I generic 122.7M+4.00I
G5 p=17159: fast = generic on 800/800 pairs (0 degenerate) | add fast 41.0M+1.00I generic 113.0M+4.00I | double fast 51.0M+1.00I generic 123.0M+4.00I
G5 p=265871: fast = generic on 800/800 pairs (0 degenerate) | add fast 41.0M+1.00I generic 113.0M+4.00I | double fast 51.0M+1.00I generic 123.0M+4.00I
```

glue_summary.txt

```
H2 bits 28 h 1 Jglue: steps/p 0.4353 (s.e. 0.0168, 40 runs) | mean cost (I=3M, w=0) 5.0237e5 | rho theory 0.8860
H2 bits 28 h 1 R: steps/p 0.9556 (s.e. 0.0638, 40 runs) | mean cost (I=3M, w=0) 3.5065e5 | rho theory 0.8860
H2 bits 28 h 4 Jglue: steps/p 0.3561 (s.e. 0.0182, 40 runs) | mean cost (I=3M, w=0) 4.1819e5 | rho theory 0.4430
H2 bits 28 h 4 R: steps/p 0.4478 (s.e. 0.0353, 40 runs) | mean cost (I=3M, w=0) 2.0860e5 | rho theory 0.4430
H2 bits 32 h 1 Jglue: steps/p 0.4260 (s.e. 0.0184, 40 runs) | mean cost (I=3M, w=0) 1.5867e6 | rho theory 0.8860
H2 bits 32 h 1 R: steps/p 0.8440 (s.e. 0.0691, 40 runs) | mean cost (I=3M, w=0) 9.8975e5 | rho theory 0.8860
H2 bits 32 h 4 Jglue: steps/p 0.3163 (s.e. 0.0215, 40 runs) | mean cost (I=3M, w=0) 1.2304e6 | rho theory 0.4430
H2 bits 32 h 4 R: steps/p 0.4155 (s.e. 0.0308, 40 runs) | mean cost (I=3M, w=0) 5.5382e5 | rho theory 0.4430
```

```
H2 bits 36 h 1 Jglue: steps/p 0.4387 (s.e. 0.0144, 40 runs) | mean cost (I=3M, w=0) 5.9674e6 | rho theory 0.8860
H2 bits 36 h 1 R: steps/p 0.7782 (s.e. 0.0757, 40 runs) | mean cost (I=3M, w=0) 3.2959e6 | rho theory 0.8860
H2 bits 36 h 4 Jglue: steps/p 0.3372 (s.e. 0.0179, 40 runs) | mean cost (I=3M, w=0) 4.8011e6 | rho theory 0.4430
H2 bits 36 h 4 R: steps/p 0.4033 (s.e. 0.0411, 40 runs) | mean cost (I=3M, w=0) 1.8353e6 | rho theory 0.4430
H2 bits 40 h 1 Jglue: steps/p 0.4428 (s.e. 0.0158, 40 runs) | mean cost (I=3M, w=0) 2.4105e7 | rho theory 0.8860
H2 bits 40 h 1 R: steps/p 0.8600 (s.e. 0.0681, 40 runs) | mean cost (I=3M, w=0) 1.4378e7 | rho theory 0.8860
H2 bits 40 h 4 Jglue: steps/p 0.3496 (s.e. 0.0175, 40 runs) | mean cost (I=3M, w=0) 1.9498e7 | rho theory 0.4430
H2 bits 40 h 4 R: steps/p 0.4446 (s.e. 0.0314, 40 runs) | mean cost (I=3M, w=0) 7.7110e6 | rho theory 0.4430
H2 bits 28 h 1: Jglue/rho cost ratio (measured, I=3M, w=0) 1.433
H2 bits 28 h 4: Jglue/rho cost ratio (measured, I=3M, w=0) 2.005
H2 bits 32 h 1: Jglue/rho cost ratio (measured, I=3M, w=0) 1.603
H2 bits 32 h 4: Jglue/rho cost ratio (measured, I=3M, w=0) 2.222
H2 bits 36 h 1: Jglue/rho cost ratio (measured, I=3M, w=0) 1.811
H2 bits 36 h 4: Jglue/rho cost ratio (measured, I=3M, w=0) 2.616
H2 bits 40 h 1: Jglue/rho cost ratio (measured, I=3M, w=0) 1.677
H2 bits 40 h 4: Jglue/rho cost ratio (measured, I=3M, w=0) 2.529
H2 Jglue: measured per Jacobian op 41.3M + 1.00I
H2 band bits 28 h 1: Jglue/rho in [0.594, 1.514] over 12 cells (IC cheaper in 6) -> unresolved
H2 band bits 28 h 4: Jglue/rho in [0.823, 2.136] over 12 cells (IC cheaper in 3) -> unresolved
H2 band bits 32 h 1: Jglue/rho in [0.706, 1.712] over 12 cells (IC cheaper in 6) -> unresolved
H2 band bits 32 h 4: Jglue/rho in [0.972, 2.372] over 12 cells (IC cheaper in 2) -> unresolved
H2 band bits 36 h 1: Jglue/rho in [0.815, 1.938] over 12 cells (IC cheaper in 3) -> unresolved
H2 band bits 36 h 4: Jglue/rho in [1.176, 2.788] over 12 cells (IC cheaper in 0) -> rho cheaper in every cell
H2 band bits 40 h 1: Jglue/rho in [0.759, 1.795] over 12 cells (IC cheaper in 6) -> unresolved
H2 band bits 40 h 4: Jglue/rho in [1.144, 2.690] over 12 cells (IC cheaper in 0) -> rho cheaper in every cell
H2 all answers verified: true
```

host.txt

```
Architecture:                         x86_64
CPU op-mode(s):                       32-bit, 64-bit
Address sizes:                        48 bits physical, 48 bits virtual
Byte Order:                           Little Endian
CPU(s):                               32
On-line CPU(s) list:                  0-31
Vendor ID:                            AuthenticAMD
Model name:                           AMD EPYC 4564P 16-Core Processor
CPU family:                           25
Model:                                97
Thread(s) per core:                   2
Core(s) per socket:                   16
Socket(s):                            1
Stepping:                             2
CPU(s) scaling MHz:                   78%
CPU max MHz:                          5881.0000
CPU min MHz:                          400.0000
BogoMIPS:                             8983.20
Flags:                                fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ht syscall nx mmxext fxsr_opt pdpe1gb rdtscp lm
constant_tsc rep_good amd_lbr_v2 nopl nonstop_tsc cpuid extd_apicid aperfmperf rapl pni pclmulqdq monitor ssse3 fma cx16 sse4_1 sse4_2 movbe popcnt aes xsave avx f16c rdrand lahf_lm cmp_legacy
svm extapic cr8_legacy abm sse4a misalignsse 3dnowprefetch osvw ibs skinit wdt tce topoext perfctr_core perfctr_nb bpext perfctr_llc mwaitx cpb cat_l3 cdp_l3 hw_pstate ssbd mba perfmon_v2 ibrs
ibpb stibp ibrs_enhanced vmmcall fsgsbase bmi1 avx2 smep bmi2 erms invpcid cqm rdt_a avx512f avx512dq rdseed adx smap avx512ifma clflushopt clwb avx512cd sha_ni avx512bw avx512vl xsaveopt xsavec
xgetbv1 xsaves cqm_llc cqm_occup_llc cqm_mbm_total cqm_mbm_local user_shstk avx512_bf16 clzero irperf xsaveerptr rdpru wbnoinvd cppc arat npt lbrv svm_lock nrip_save tsc_scale vmcb_clean
flushbyasid decodeassists pausefilter pfthreshold avic v_vmsave_vmload vgif x2avic v_spec_ctrl vnmi avx512vbmi umip pku ospke avx512_vbmi2 gfni vaes vpclmulqdq avx512_vnni avx512_bitalg
avx512_vpopcntdq rdpid overflow_recov succor smca flush_l1d
Virtualization:                       AMD-V
L1d cache:                            512 KiB (16 instances)
L1i cache:                            512 KiB (16 instances)
L2 cache:                             16 MiB (16 instances)
L3 cache:                             64 MiB (2 instances)
NUMA node(s):                         1
NUMA node0 CPU(s):                    0-31
Vulnerability Gather data sampling:   Not affected
Vulnerability Itlb multihit:          Not affected
Vulnerability L1tf:                   Not affected
```

```
Vulnerability Mds:                    Not affected
Vulnerability Meltdown:               Not affected
Vulnerability Mmio stale data:        Not affected
Vulnerability Reg file data sampling: Not affected
Vulnerability Retbleed:               Not affected
Vulnerability Spec rstack overflow:   Mitigation; Safe RET
Vulnerability Spec store bypass:      Mitigation; Speculative Store Bypass disabled via prctl
Vulnerability Spectre v1:             Mitigation; usercopy/swapgs barriers and __user pointer sanitization
Vulnerability Spectre v2:             Mitigation; Enhanced / Automatic IBRS; IBPB conditional; STIBP always-on; RSB filling; PBRSB-eIBRS Not affected; BHI Not affected
Vulnerability Srbds:                  Not affected
Vulnerability Tsx async abort:        Not affected
               total        used        free      shared  buff/cache   available
Mem:             124          27          81           0          18          97
Swap:              0           0           0
16
Linux c7864259a52b 6.8.0-45-generic #45~22.04.1-Ubuntu SMP PREEMPT_DYNAMIC Wed Sep 11 15:25:05 UTC 2 x86_64 GNU/Linux
rustc 1.99.1 (58cf8f9c3 2026-10-13)
```

rusage.txt

```
Command being timed: "bash -c mkdir -p examples && curl -sSfL https://raw.githubusercontent.com/aburan28/crypto/0bd2a0f5c0b0f0005d57850c12466fa82b414c59/examples/n2_routes.rs -o examples/
n2_routes.rs && echo 'fd8faeb6fb42db70933592f2dd2ad7d4ceecb66900b738e45e42f76ede88d66e  examples/n2_routes.rs' | sha256sum -c - && rustc --edition 2021 -O examples/n2_routes.rs -o n2 && ./n2
"$OUT" gluecheck && N2_THREADS=16 ./n2 "$OUT" glue"
  User time (seconds): 78.58
  System time (seconds): 2.09
  Percent of CPU this job got: 563%
  Elapsed (wall clock) time (h:mm:ss or m:ss): 0:14.31
  Average shared text size (kbytes): 0
  Average unshared data size (kbytes): 0
  Average stack size (kbytes): 0
  Average total size (kbytes): 0
  Maximum resident set size (kbytes): 627408
  Average resident set size (kbytes): 0
  Major (requiring I/O) page faults: 39
  Minor (reclaiming a frame) page faults: 674573
  Voluntary context switches: 1497
  Involuntary context switches: 1338
  Swaps: 0
  File system inputs: 0
  File system outputs: 13192
  Socket messages sent: 0
  Socket messages received: 0
  Signals delivered: 0
  Page size (bytes): 4096
  Exit status: 0
```

glue.csv (first 12 of 640 rows)

bits	h	instance	p	N	ell	twist	target	arm	steps	relations	jac_ops	walk_M	walk_I	walk_native	jac_M	jac_I	la_native	eadd_M	eadd_I	verified	seconds
28	1	0	17317	299862469	299862469	0	0	Jglue	4639	2318	7385	337941	7367	13914	318815	7329	15843	12	1	1	0.002
28	1	0	17317	299862469	299862469	0	0	R	20634	0	0	336477	26381	41200	0	0	0	12	1	1	0.004
28	1	0	17317	299862469	299862469	0	1	Jglue	6629	3356	9588	437549	9572	19971	410279	9534	27095	12	1	1	0.003
28	1	0	17317	299862469	299862469	0	1	R	28472	0	0	435680	34550	56983	0	0	0	12	1	1	0.005
28	1	0	17317	299862469	299862469	0	2	Jglue	9763	4820	12673	576321	12675	29167	536850	12631	48101	12	1	1	0.004
28	1	0	17317	299862469	299862469	0	2	R	19398	0	0	321035	25130	38815	0	0	0	12	1	1	0.003
28	1	0	17317	299862469	299862469	0	3	Jglue	3755	1876	6503	298280	6492	11263	282678	6454	12117	12	1	1	0.002
28	1	0	17317	299862469	299862469	0	3	R	10060	0	0	205155	15550	20108	0	0	0	12	1	1	0.002
28	1	0	17317	299862469	299862469	0	4	Jglue	9036	4542	11879	540388	11859	27156	503557	11821	42555	12	1	1	0.004

28	1	0	17317	299862469	299862469	0	4	R	13843	0	0	252276	19433	27687	0	0	0	12	1	1	0.003
28	1	0	17317	299862469	299862469	0	5	Jglue	9193	4585	12130	551577	12109	27556	514451	12084	43635	12	1	1	0.004
28	1	0	17317	299862469	299862469	0	5	R	5287	0	0	145253	10614	10573	0	0	0	12	1	1	0.001

Graph

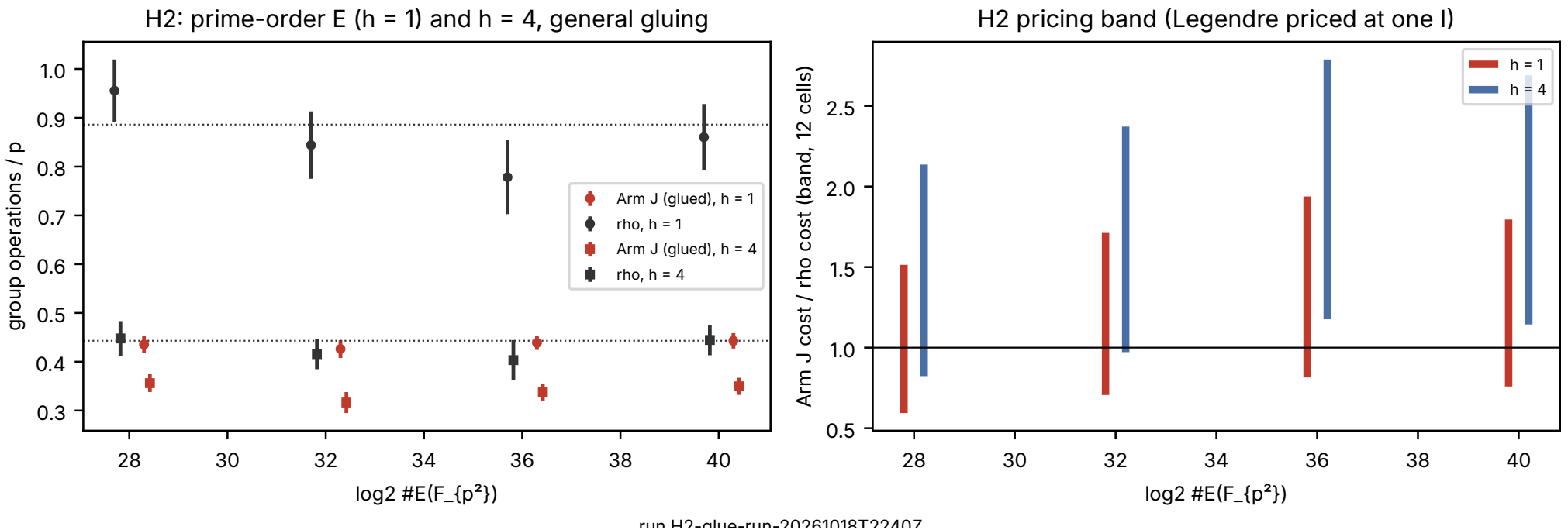


Figure 1: H2: general E E^sigma gluing, h = 1 and 4: group operations per p and the pricing band.